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Age of the Firesand River carbonatite complex from paleomagnetism[1]

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Received February 7, 1989

Revision accepted April 12, 1989

The 2.3 km diameter Firesand River complex intrudes Archean volcanics and granites of the Wawa Subprovince in the Superior Province about 8 km east of Wawa, Ontario. It has given differing Middle Proterozoic K–Ar biotite ages of 1018 ± 50 and 1097 Ma. Alternating-field and thermal step demagnetization of specimens from three calcitic carbonatite sites, five ferruginous dolomitic carbonatite sites, and one lamprophyre dike site isolated a stable mean direction of 290°, 33° (α_{95} = 12°). Isothermal remanent magnetization tests indicate the remanence is held by single- to pseudosingle-domain magnetite and hematite in the carbonatite. The dike remanence is Keweenawan in age, thereby confirming its genetic relationship to the complex, and it gives a positive partial contact test with its host rock, indicating no postintrusive remagnetization. The blocking-temperature spectra indicate that postintrusive uplift has not exceeded about 4 km. The pole position for the complex is 183°E, 27°N (dp = 8°, dm = 13°). This pole lies directly on the well-dated Keweenawan apparent polar wander path, giving an age of 1090 ± 10 Ma, in agreement with the older K–Ar age. It also confirms geologic and aeromagnetic evidence that the complex has not been tectonically tilted since emplacement.

Le complexe de Firesand River, ayant 2,3 km de diamètre, recoupe les volcanites et les granites archéens de la sous-province de Wawa dans la province du Lac Supérieur, à environ 8 km à l'est de Wawa, Ontario. Les roches de ce complexe protérozoïque moyen ont fourni des dates K–Ar sur biotites passablement différentes, soit 1018 ± 50 et 1097 Ma. La désaimantation par champ alternatif et par chauffage de trois spécimens de carbonatite calcitique, de cinq spécimens de carbonatite dolomitique et ferrugineuse et d'un spécimen d'un dyke de lamprophyre ont isolé une direction moyenne stable de 290°, 33° (α_{95} = 12°). Les tests des inversions indiquent que la rémanence est portée par les monodomaines ou pseudo-monodomaines de magnétite et d'hématite contenues dans la carbonatite. La rémanence du dyke est d'âge keweenawien; par conséquent il est génétiquement associé au complexe, et le test de contact avec sa roche encaissante étant en partie positif témoigne de l'absence d'une réaimantation postintrusive. Les spectres des températures de blocage indiquent que le soulèvement postintrusif était inférieur à 4 km. La position du pôle du complexe est 183°E, 27°N (dp = 8°, dm = 13°). Ce pôle tombe directement sur le tracé de la courbe de dérive apparente des pôles pour le Keweenawien et fournit un âge de 1090 ± 10 Ma, correspondant au plus vieux des âges K–Ar. Cet âge supporte les déductions géologiques et aéromagnétiques attestant l'absence d'un basculement tectonique du complexe depuis sa mise en place.

[Traduit par la revue]

Can. J. Earth Sci. **26**, 2401–2405 (1989)

Introduction

The Firesand River complex is located 8 km due east of Wawa, Ontario (Fig. 1). Gittens *et al.* (1967) reported K–Ar biotite ages of 1008 and 1087 Ma for the complex, so 1048 Ma is commonly cited as its age (Woolley 1987). Except for a few thin lamprophyre dikes of <20 cm width (Sage 1983), exposure is limited to a few small, moss-covered carbonatite outcrops. Although these rocks are not well suited to conventional radiometric dating, they are good paleomagnetic recorders.

Geology

The Firesand River complex is a slightly elliptical stock of about 2.3 km in diameter in the Wawa Subprovince of the Superior Province of the Canadian Shield (Fig. 2). Its geology is known primarily from drilling for niobium, apatite, and limestone deposits (Parsons 1961; Sage *et al.* 1982; Sage 1983). It intrudes Archean greenstones and granites of the Wawa greenstone belt, and these have been fenitized up to 1000 m from the contact of the complex. Its perimeter phase of calcitic carbonatite includes sovite and silicocarbonatite that contains abundant biotite, apatite, magnetite, pyrrhotite, and pyrochlore, giving a strong circular aeromagnetic anomaly

[1]Lithoprobe Publication 93.

Printed in Canada / Imprimé au Canada

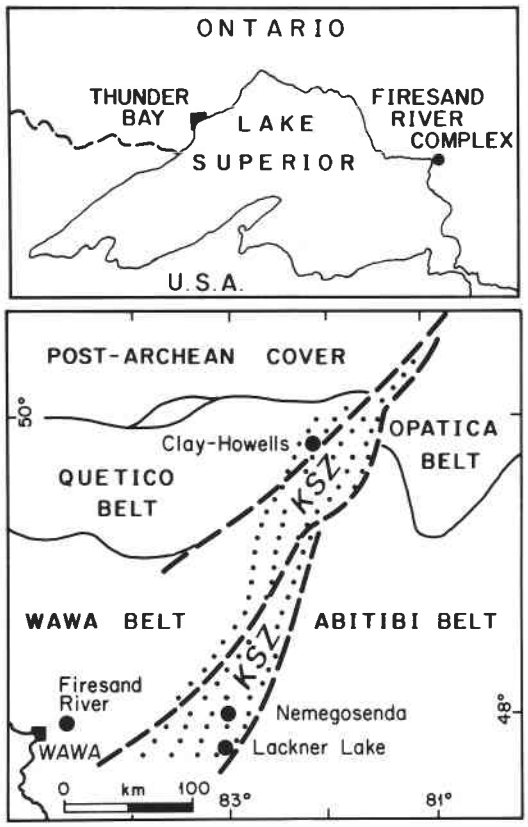


FIG. 1. Location map for the Firesand River complex. The stippled area is the Kapuskasing structural zone (KSZ); the locations of some other alkalic carbonatite complexes are shown.

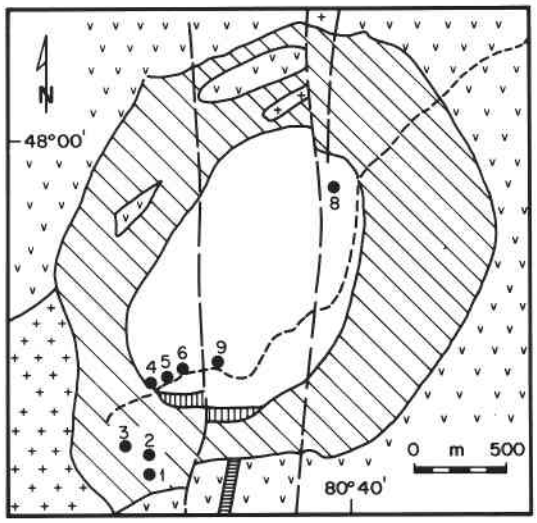


FIG. 2. Site locations and geology of the Firesand River complex. (Site 7 is in a rockcut on the south side of Highway 101 about 300 m west of where it crosses the Chabanel–Esquega township boundary or 3.4 km due north of site 8 (Sage *et al.* 1981).) Patterns in the complex: diagonal lines, calcitic carbonatite; no pattern, dolomitic carbonatite; vertical lines, mafic carbonatite. Patterns in the Archean host rocks: vees, greenstone; crosses, granite; horizontal lines, Matachewan diabase dike.

of about 1.0 A m^{-1} that defines the outer contact (ODM–GSC 1963*a*, 1963*b*). This perimeter phase also has a strong concentric foliation that dips steeply inwards. The core of the complex is about 1 km in diameter and comprises ferriginous rauhaugite with minor quartz and barite. Narrow carbonatite and lamprophyre dikes cut the complex and probably radiate from it. The circular shape, concentric petrology, foliation attitudes, and magnetic anomaly all indicate that the complex was emplaced vertically and has not subsequently been tilted. Parsons (1961) interpreted minor faults based on surface magnetics, but Sage (1983) found no proof of these faults. There is no evidence of any postintrusive regional metamorphism of the complex. Using current decay constants, Easton (1986) recalculated the K–Ar ages of Gittins *et al.* (1967) at 1018 ± 50 and 1097 Ma.

Methods

Three calcitic and five dolomitic carbonatite outcrops and one lamprophyre dike outcrop were found. Three to six oriented cores were drilled from each of these meager, moss-covered outcrops, giving nine sites in total (Fig. 2). The dike at site 7, about 3 km north of the complex, was too narrow to sample for a full paleomagnetic contact test. Measurements were made with an automated Canadian Thin Films cryogenic magnetometer, a Sapphire Instruments SI-4 alternating-field (AF) demagnetizer and direct-current (DC) magnetizer, and a Schonstedt TSD-1 thermal demagnetizer. Two pilot specimens from each site were AF demagnetized in 13 steps to 160 mT, and a further two specimens were thermally demagnetized in 15 steps to 680°C. The remaining specimens were demagnetized in three AF steps and one final thermal step based on the pilot specimen results for each site. Isothermal remanent magnetization (IRM) tests up to 200 mT were run on six typical specimens.

Results

The natural remanent magnetization (NRM) intensities of the carbonatite and lamprophyre specimens are relatively weak and in the 10^{-4}–10^{-3} A m^{-1} range. On AF step demagnetization, four of the carbonatite sites (1, 2, 6, and 8) show a single, very stable remanence component, two sites (3 and 4) show the initial removal of small viscous remanent magnetization (VRM) components with an underlying stable remanence, and the remaining two sites (5 and 9) show the initial removal of large VRM components with an underlying stable remanence (Figs. 3*a*–3*c*). The lamprophyre dike specimens undergo rapid AF demagnetization with lower directional stability (Fig. 3*d*). On thermal step demagnetization a single, very stable remanence component is found at four sites. Three of these sites (5, 6, and 8) have blocking temperatures predominantly in the 600°–675°C range, characteristic of hematite, and one site (1) has blocking temperatures in the 500°–585°C range, characteristic of magnetite (Figs. 4*a*, 4*b*). The remaining four carbonatite sites have significant initial VRM components with an underlying quite stable remanence residing in magnetite and (or) hematite (Figs. 4*c*, 4*d*). The lamprophyre dike specimens show a rapid remanence intensity decay to about 1% by 450°C, with erratic directional changes thereafter.

Six representative specimens were given an IRM in seven DC steps up to 200 mT. The IRM-acquisition curves (Fig. 5*a*) show that the lamprophyre dike specimen from site 7 saturates

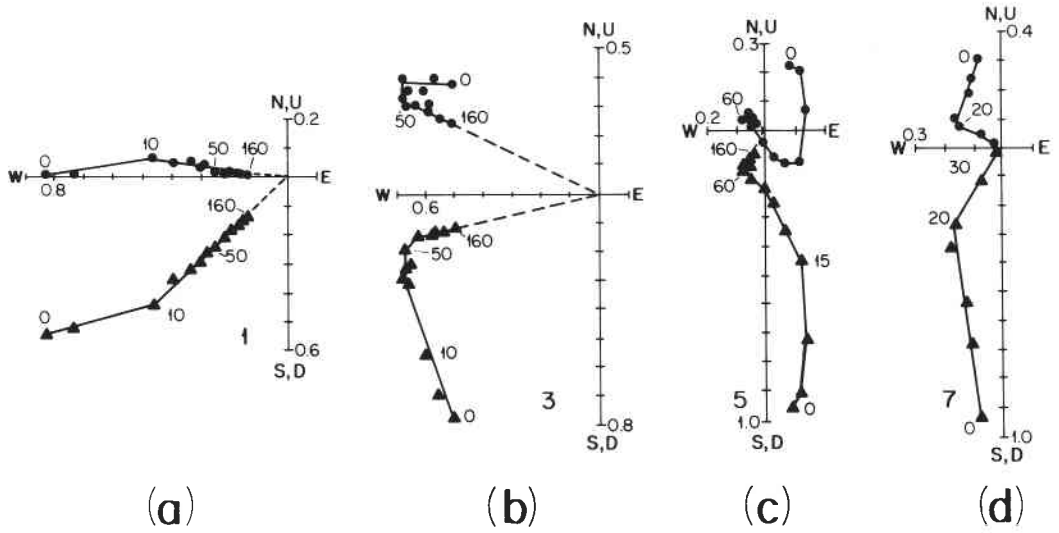


FIG. 3. Orthogonal-vector AF-demagnetization diagrams for specimens from sites 1 (*a*), 3 (*b*), 5 (*c*), and 7 (*d*). Horizontal (N, E, S, W,) and vertical (W, U, E, D) projections are shown by circles and triangles, respectively. Axial intensities are expressed as a ratio of the NRM intensity. The step intensities are in milliteslas.

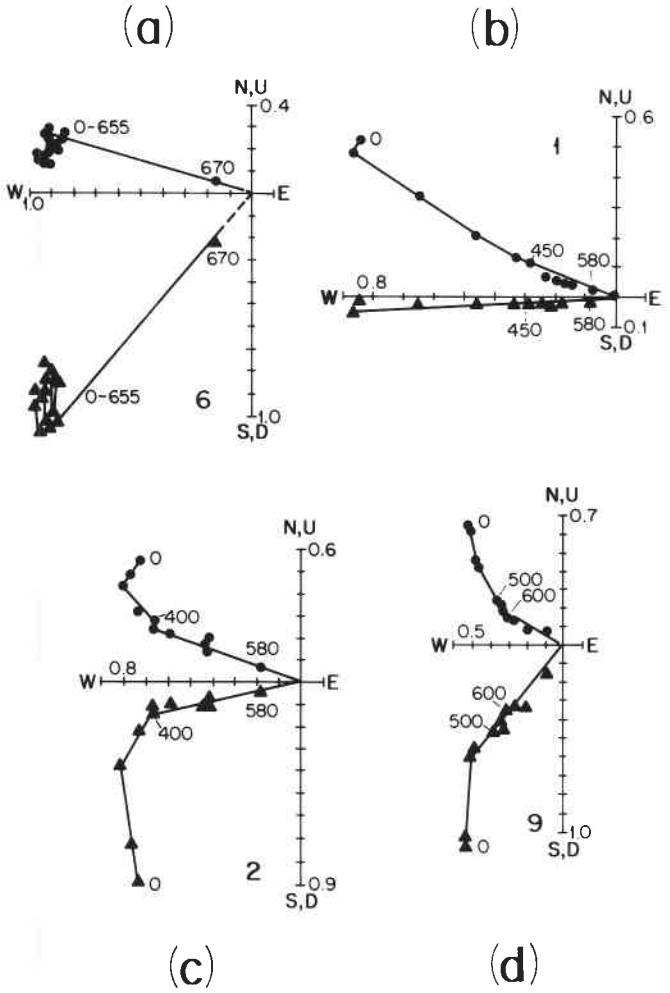


FIG. 4. Orthogonal-vector thermal-demagnetization diagrams for specimens from sites 6 (*a*), 1 (*b*), 2 (*c*), and 9 (*d*). Plotting conventions are as in Fig. 3, except the step intensities are in degrees Celsius.

by 200 mT, indicating magnetite, the ferruginous dolomitic carbonatite from site 8 shows a linear increase, indicating hematite, and the remaining four carbonatite specimens have intermediate curves, indicating both magnetite and hematite in varying proportions. The specimens were then AF demagnetized in six steps to 40 mT (Fig. 5*b*). The lamprophyre dike specimen shows the dominance of pseudosingle-domain (PSD) with some multidomain (MD) magnetite. The magnetite in the carbonatite specimens falls between the PSD and single-domain (SD) bounds.

For each specimen, three or more demagnetization step directions were averaged to get its mean direction. The specimen directions are very well clustered, so a more sophisticated analysis would not significantly change the site-mean directions. The directions of the specimens were averaged to get the core-mean directions, and these were averaged in turn to get the site-mean direction (Fisher 1953). Two sites of carbonatite (1 and 3) had one aberrant core direction each, which might well have been caused by unnoticed block rotations in the poor outcrop exposures. One core from the lamprophyre dike site (7) did not yield a stable direction. These three core-mean directions were omitted from the analysis (Table 1). The site-mean direction for the lamprophyre dike is in the middle of those from the carbonatite complex itself, indicating that they were coeval (Fig. 6). The mafic volcanic host rocks at the dike site give a different remanence direction of about $D = 355°$, $I = 35°$, thereby indicating in a partial contact test that the dike has not been remagnetized by metamorphism since emplacement (D. T. A. Symons, unpublished data). It is evident that the stable remanence in the nine sites is held by both magnetite and hematite over a broad range of coercivities and blocking temperatures. There is no evidence of an underlying older remanence component. Also, no significant difference was observed between the calicitic and dolomitic carbonatite specimens or sites. The resulting mean remanence direction for all nine sites in the Firesand River complex is $D = 290°$, $I = 33°$ ($\alpha_{95} = 12°$). This direction gives a paleopole position of 177°W, 27°N (dp = 8°, dm = 13°). Excluding the

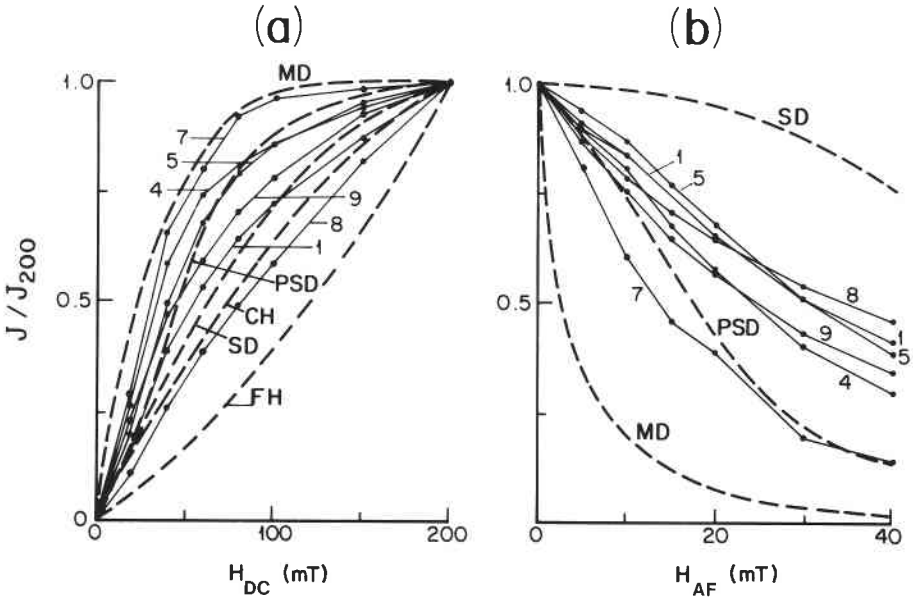


FIG. 5. IRM data showing (*a*) acquisition of IRM up to 200 mT and (*b*) AF demagnetization of the IRM acquired at 200 mT. The type curves for multidomain (MD), pseudosingle-domain (PSD), and single-domain (SD) magnetite and for coarse-grained (CH) and fine-grained (FH) hematite are adapted from Dunlop (1971, 1972, 1973, 1981). J/J_{200} is the ratio of the measured remanence intensity acquired at 200 mT. H_{DC} and H_{AF} are the intensities of the magnetizing and demagnetizing fields, respectively.

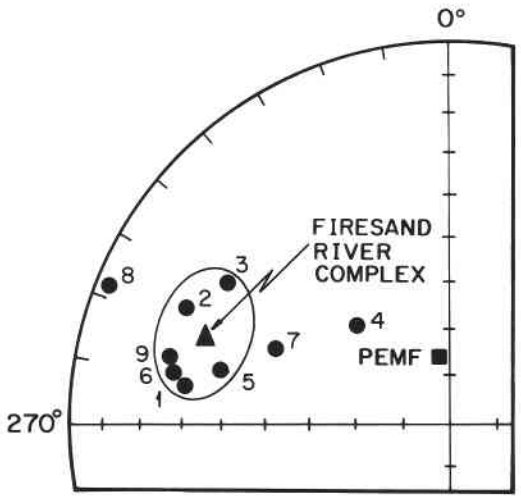


FIG. 6. Site-mean remanence directions plotted on the lower hemisphere of an equal-area stereogram. The triangle indicates the mean direction for the Firesand River complex and is surrounded by its cone of 95% confidence. The square shows the present Earth's magnetic field (PEMF) direction.

TABLE 1. Mean remanence statistics

Site	Number of cores, specimens	Mean remanence direction D (°)	I (°)	α_{95} (°)	k	Demagnetization range represented (mT, °C)
1	4, 10	278	29	14	45	10–160, 525–560
2	5, 14	294	26	11	68	10–160, 400–650
3	4, 8	303	32	12	58	15–160, 500–540
4	6, 9	317	61	19	13	20–160, 450–650
5	6, 9	283	38	19	13	20–160, 500–670
6	6, 7	281	27	20	12	15–160, 100–670
7	4, 6	293	49	27	12	20–100, 200–560
8	5, 10	292	4	11	52	10–160, 100–670
9	3, 6	283	26	53	6	15–160, 100–670
Mean	(9 sites)	290.2	32.8	11.9	19.7	
Mean	(7 sites)[a]	287.6	32.7	8.5	50.9	

NOTES: The site-mean directions were computed with two-tiered statistics (Fisher 1953) and are defined by their declination (D), inclination (I), radius of the cone of 95% confidence (α_{95}) and precision parameter (k). The last column gives the range of AF and thermal demagnetization treatments used to isolate the specimen directions. The mean-site location is 84.67°W, 47.99°N.

[a]Excludes sites 4 and 8.

two most aberrant site means (sites 4 and 8) improves the precision but does not change the pole position significantly.

Discussion

If a magma temperature for the complex of about 800°C (LeBas 1977) and a host-rock temperature of about 100°C are assumed, the core of the complex took about 2×10^5 years to cool to about 500°C or through the blocking temperatures of the stable magnetite domains (Carslaw and Jaeger 1959). This is about an order of magnitude greater time period than is required to average out the effects of paleosecular variation on the remanence direction to within about 2° (Champion 1980), indicating that a representative paleomagnetic pole has been isolated.

The pole position for the Firesand River complex plots on the Keweenawan apparent polar wander path (APWP) for North America, where it is well defined both paleomagnetically and radiometrically (Halls and Pesonen 1982; Palmer and Davis 1987; Symons 1989) (Fig. 7). The pole indicates an age of 1090 ± 10 Ma for the complex, in close agreement with the older K–Ar biotite age of 1097 Ma (Gittins *et al.* 1967; Easton 1986). It also confirms the geologic and aeromagnetic evidence that the complex has not been tilted since emplacement. The Firesand polarity is normal, like those found in the

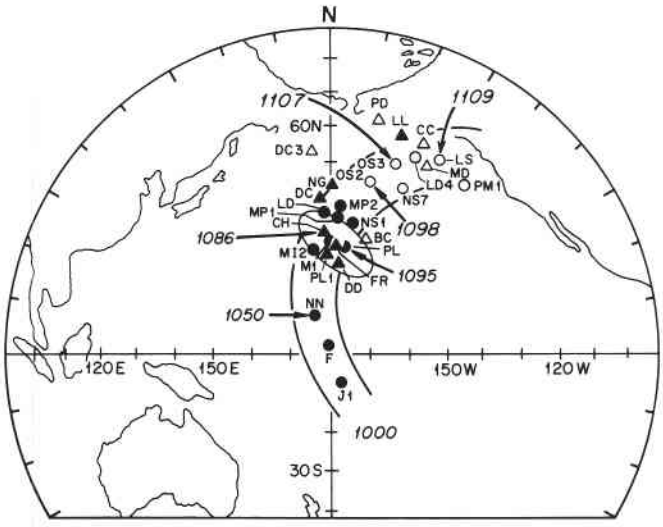


FIG. 7. APWP for Keweenawan time from Halls and Pesonen (1982) and Palmer and Davis (1987). The added poles are for the Nemegosenda (NG) (Symons and Garber 1974), Clay–Howells (CH) (Lewchuk and Symons 1988), Lackner Lake (LL) (Symons 1989), and Firesand River (FR, this study) complexes. The oval of 95% confidence is shown for the Firesand River complex. The radiometric ages are in millions of years.

Nemegosenda, Lackner Lake, and Clay–Howells complexes, which are definitely in the Kapuskasing structural zone (KSZ) (Fig. 1). The Nemegosenda complex gives a U–Pb zircon age of 1107^{+6}_{-4} Ma; Lackner Lake, a Rb–Sr whole-rock isochron age of 1095 ± 30 Ma; Clay–Howells, a Rb–Sr whole-rock isochron age of 1075 ± 15 Ma (Bell *et al.* 1982; Heaman *et al.* 1988). These data support the suggestion of Sage (1983) that the Firesand River complex was emplaced in faults of the KSZ system.

Finally, the Keweenawan remanence direction is seen in magnetic domains that are blocked at ≤300°C in both carbonatite and lamprophyre specimens (e.g., Figs. 4*b*, 4*c*). If a geothermal gradient of 25°C km^{-1} at 1.1 Ga is assumed, this indicates that the pluton was emplaced into host rock with an ambient temperature of <100°C or within about 4 km of surface (Blackwell 1971; Cermak and Jessop 1971; Pullaiah *et al.* 1975).

Acknowledgments

I wish to thank E. D. Frey of the Ontario Ministry of Northern Development and Mines for leading us to most of the outcrops. M. T. Lewchuk, J. A. Ceman, W. Clinton, and J. Garber assisted in the collection, preparation, and measurement of the samples. I also wish to thank E. Irving and H. C. Palmer for their useful suggestions for improving the manuscript. This work was funded by the Natural Sciences and Engineering Research Council of Canada through a Lithoprobe grant.

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